AI vs Neural Networks

Transformer is a neural network architecture that is extremely good at understanding (very long) sequences of data. This makes it ideal for NLP (Natural Language Processing) among other things.

Shortcomings?

- Modern models are huge . . . and they are retrained infrequently
- They "hallucinate" . . . and they have no memory
- They have a fixed *context window*

How to fix this (and make "AI")?

- Maintain a memory of exchanges
- Implement **tools** that augment the model's knowledge
- Fill the context window with relevant knowledge using **RAG**

Ollama Tool Calling

In tool calling for an LLM, the model ...

- Identifies a task it cannot perform on its own (e.g., fetching real-time weather, calculating complex data, or fetching information from an API).
- Delegates the task to an external tool to get the required data or perform the action.
- Uses the results from the tool to generate a response for the user.

To define tools in Ollama, you provide ...

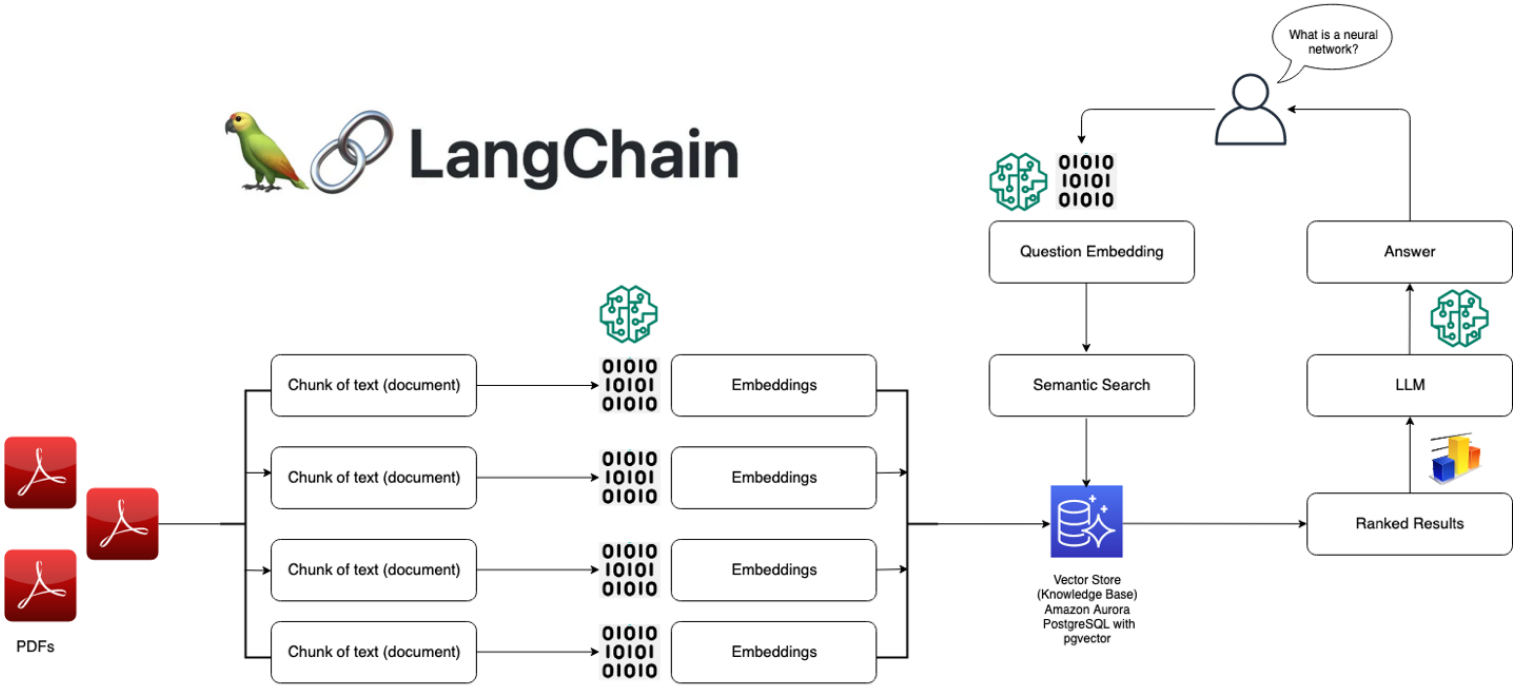
- Tool name
- Input/output format
- Functionality description

Some examples ...

Langchain

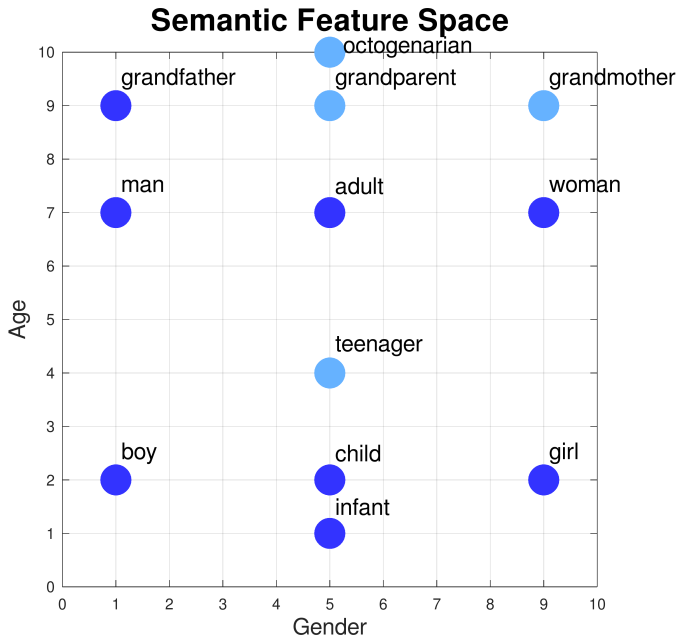
Langchain is middleware that "allows developers to connect LLMs to external data sources, manage prompt templates, and chain together multiple components for complex workflows" using

- LLM Wrappers
- Prompt Templates
- Document Loaders
- Retrieval Agents



Vector Embeddings

Idea: "Vector embedding models are algorithms that convert data (like text, images, or graphs) into numerical representations (vectors) to capture semantic meaning."



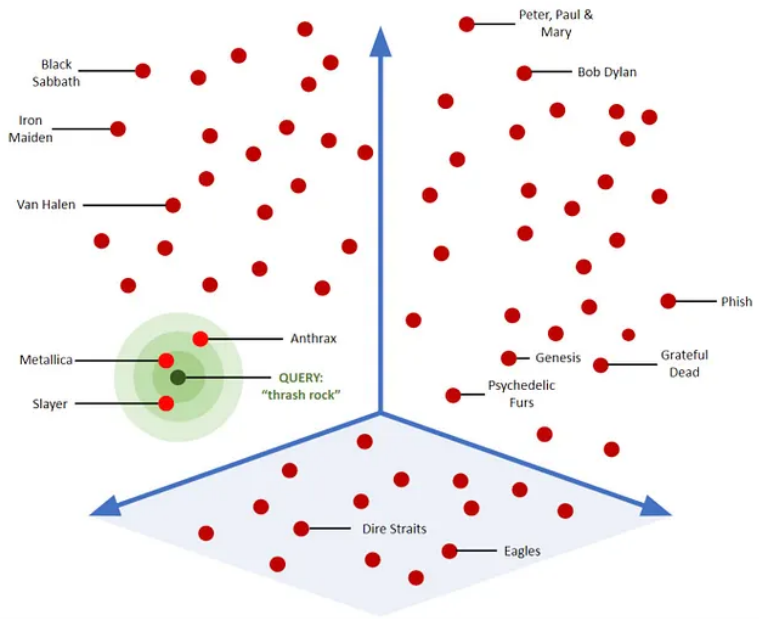
Many of these are based on derivatives of BERT (SBERT) or GPT

We can measure *similarity* using (e.g.) cosine similarity or Euclidean distance.

Vector Search

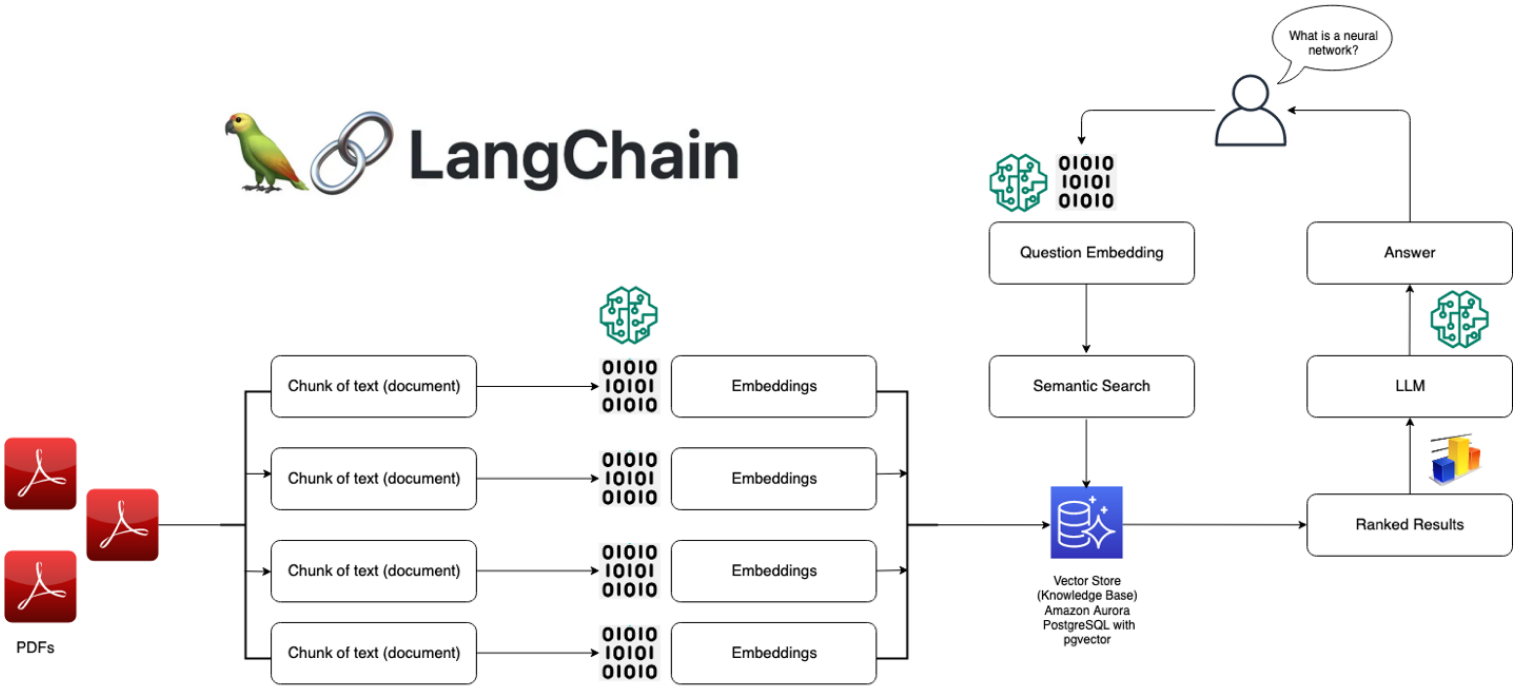
Process:

- Documents are embedded in a **vector store** through a process called **indexing**
- A search query is converted into a vector using the same embedding model
- The system calculates the similarity between the query vector and all (or a subset of) document vectors.
- Results are ranked based on their similarity scores.



Retrieval Augmented Generation

Idea: The results of this search are fed into the prompt in an LLM exchange.



RAG is a three-step pipeline consisting of: ingestion, retrieval, and generation.

An example . . .